

Assessment	: Assignment	Duration	: N/A
Module	: BWIB817 / STA843	Due date	: 15/05/26
Examiner	: Dr Brydon / Prof Smit	Time	: N/A

Worker scheduling/planning

You have recently been employed at a company and your first task in your new role is to assist with determining the assignment of workers to tasks.

The main activity of the company is the manufacturing of a specific product across various machines/tasks, but your project only includes the scheduling of workers against the tasks needed to manufacture this product i.e. the production itself is out of scope for this project.

Each task requires a specific skill set and therefore, workers can only complete tasks for which they have undergone training (these are indicated in the *tasks* table as binary indicators). Additionally, each task requires a specific number of hours to be completed and there is a “demand” for the number of times each task must be completed in a given week.

Due to the nature of the work, some workers are not available on all days in the week (indicated in the *availability* table as binary indicators for each day). Working hours are furthermore limited to a total of 8 hours per day per worker and a total of 320 hours per week for all workers.

Furthermore, workers are remunerated at varying rates, and this therefore must be taken into consideration. Your objective is therefore to minimise the cost associated with assigning a worker to a task(s).

In addition to the data sets given, your manager provides you with the below information:

- Workers are not permitted to complete more than one task per day.
- As the problem is a scheduling/planning problem, your manager requires a table as part of your solution, indicating which worker was assigned to a task(s) so that the company can efficiently plan the coming week (the output here should be for all tasks in the coming week and therefore does not need to include which day the task should be completed).

Submission details:

- The mathematical formulation and SAS implementation.
- Remember to define all variables, mathematical symbols, etc. clearly. Include descriptions/explanations for the constraints.